Delivery Delay Prediction in Logistics Systems

"Forest" Results

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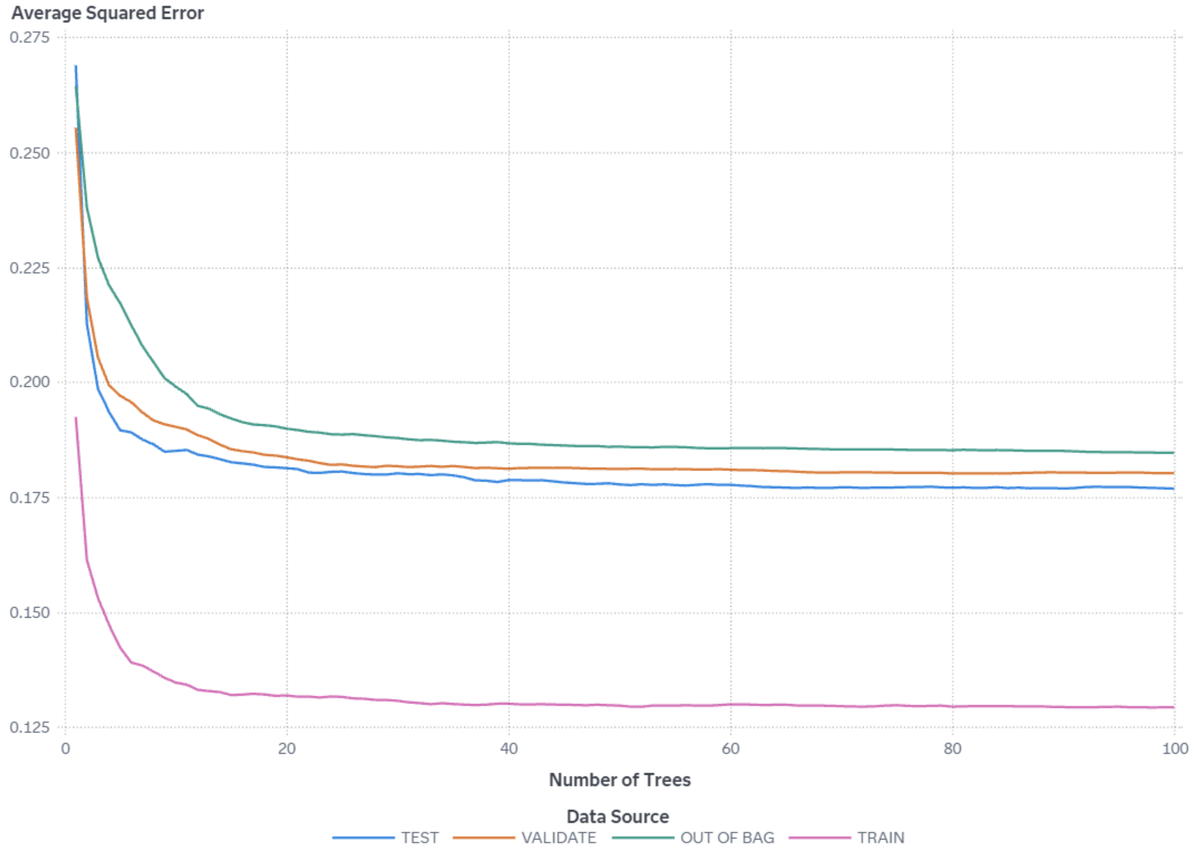
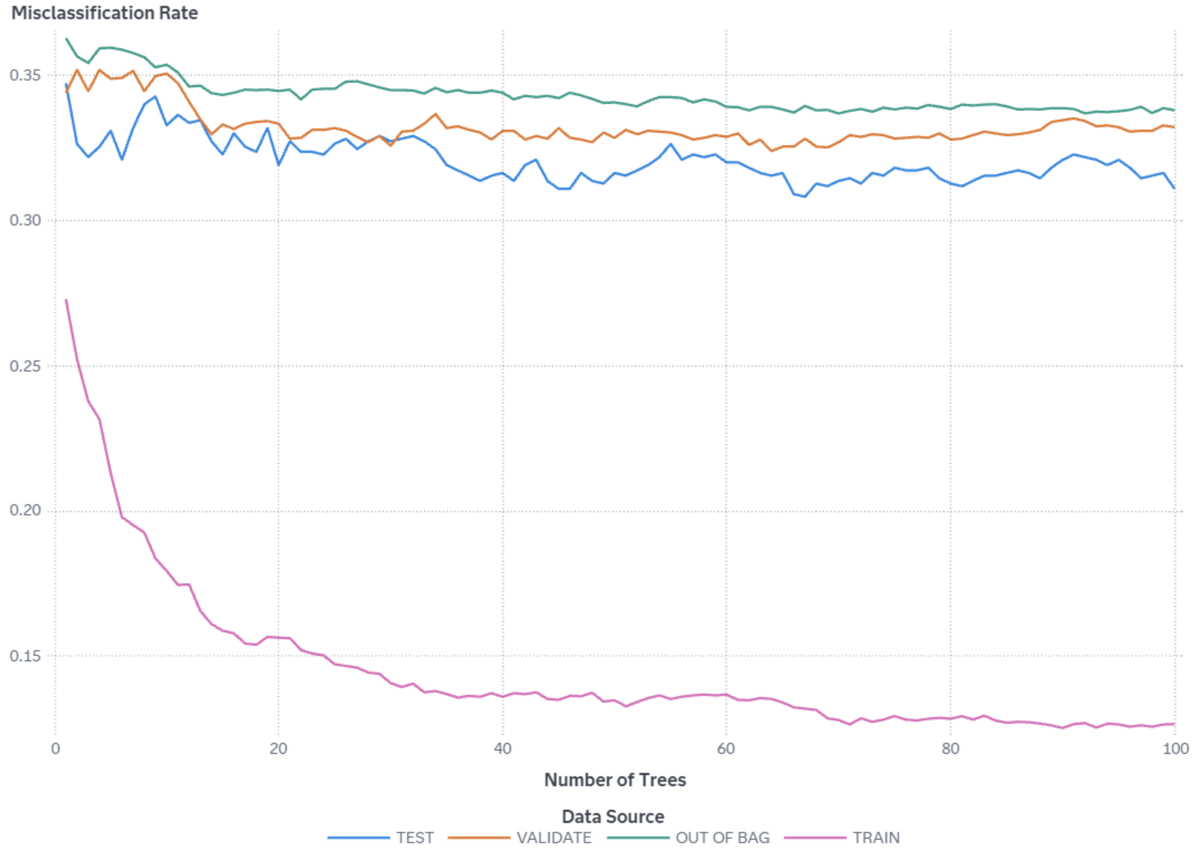
Contents

Average Squared Error



This plot shows how the average squared error changes as the number of trees in the forest increases. The training error typically decreases as the number of trees increases, but the error for the VALIDATE partition gives you an indication of how well your model generalizes. For this model, the minimum error for the VALIDATE partition is 0.18 and occurs for 80 trees.

Misclassification Rate



This plot shows how the misclassification rate changes as the number of trees in the forest increases. The training error typically decreases as the number of trees increases, but the error for the VALIDATE partition gives you an indication of how well your model generalizes. For this model, the minimum error for the VALIDATE partition is 0.324 and occurs for 64 trees.

Variable Importance

Variable Label	Role	Variable Name	Training Importance	Importance Standard Deviation	Relative Importance
	INPUT	Discount_offered	325.5808	129.3544	1
	INPUT	Weight_in_gms	245.6604	117.6634	0.7545
	INPUT	Cost_of_the_Product	132.7815	30.3922	0.4078
	INPUT	Prior_purchases	91.2513	38.5608	0.2803
	INPUT	Warehouse_block	64.9953	13.9537	0.1996
	INPUT	Customer_rating	63.4539	14.9706	0.1949
	INPUT	Customer_care_calls	59.8808	17.9286	0.1839
	INPUT	Mode_of_Shipment	26.2470	7.6626	0.0806
	INPUT	Product_importance	24.9246	7.5247	0.0766
	INPUT	Gender	17.3740	6.9544	0.0534

Score Inputs

Name	Role	Variable Level	Type
Cost_of_the_Product	INPUT	INTERVAL	N
Customer_care_calls	INPUT	NOMINAL	N
Customer_rating	INPUT	NOMINAL	N
Discount_offered	INPUT	INTERVAL	N
Gender	INPUT	NOMINAL	C
Mode_of_Shipment	INPUT	NOMINAL	C
Prior_purchases	INPUT	NOMINAL	N
Product_importance	INPUT	NOMINAL	C
Warehouse_block	INPUT	NOMINAL	C
Weight_in_gms	INPUT	INTERVAL	N

Variable Type	Variable Label	Variable Format	Variable Length
double			8
double			8
double			8
double			8
varchar			1
varchar			6
double			8
varchar			6
varchar			1
double			8

Score Outputs

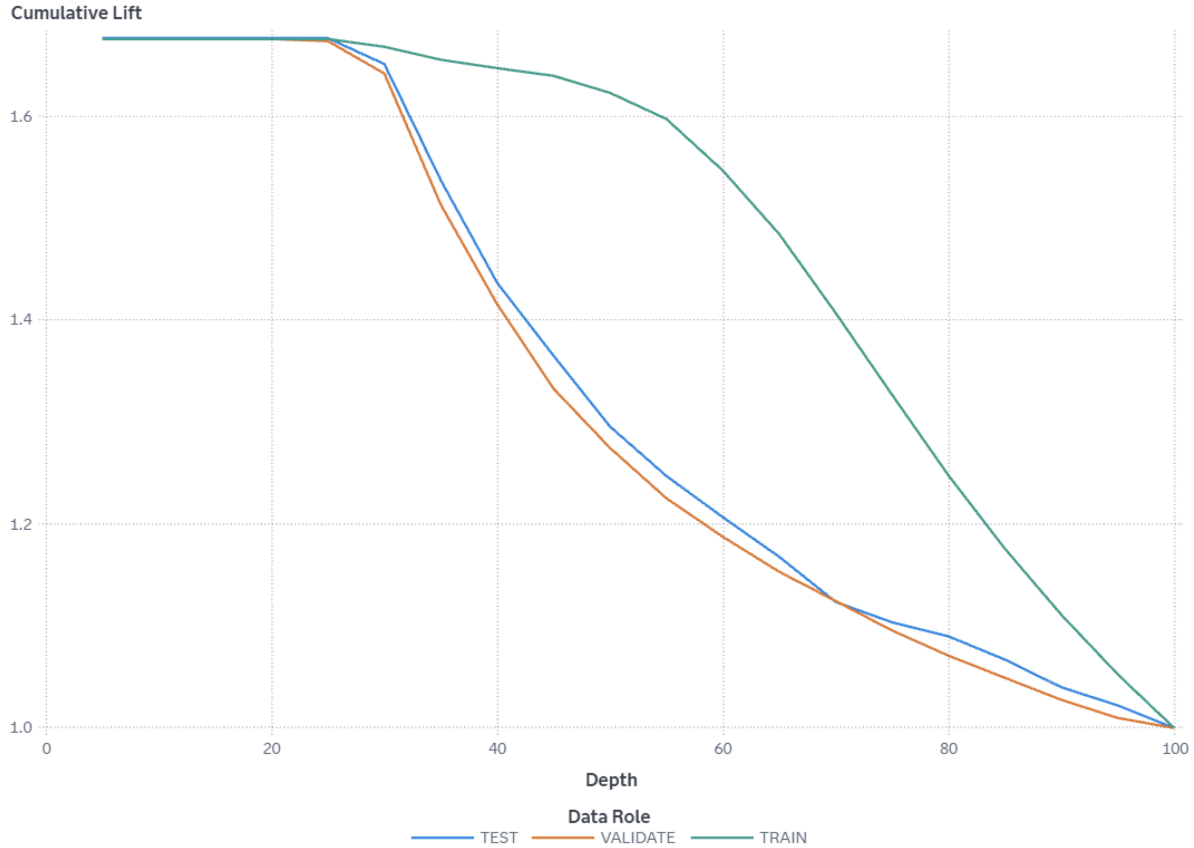
Name	Role	Type	Variable Type
EM_CLASSIFICATION	CLASSIFICATION	C	char
EM_EVENTPROBABILITY	PREDICT	N	double
EM_PROBABILITY	PREDICT	N	double
I_Reached_on_Time_Y_N	CLASSIFICATION	C	char
P_Reached_on_Time_Y_N0	PREDICT	N	double
P_Reached_on_Time_Y_N1	PREDICT	N	double
WARN	ASSESS	C	char

Variable Label	Variable Format	Variable Length	Creator
Predicted for Reached.on.Time_Y.N		12	forest
Probability for Reached.on.Time_Y.N=1		8	forest
Probability of Classification		8	forest
Into: Reached.on.Time_Y.N		12	forest
Predicted: Reached.on.Time_Y.N=0		8	forest
Predicted: Reached.on.Time_Y.N=1		8	forest
Warnings		4	forest

Function	Creator GUID
CLASSIFICATION	4ef88e4b-473f-47e

Function	Creator GUID
	7- ac02-8d184a53de d9
PREDICT	4ef88e4b-473f-47e 7- ac02-8d184a53de d9
PREDICT	4ef88e4b-473f-47e 7- ac02-8d184a53de d9
CLASSIFICATION	4ef88e4b-473f-47e 7- ac02-8d184a53de d9
PREDICT	4ef88e4b-473f-47e 7- ac02-8d184a53de d9
PREDICT	4ef88e4b-473f-47e 7- ac02-8d184a53de d9
ASSESS	4ef88e4b-473f-47e 7- ac02-8d184a53de d9

Cumulative Lift



The VALIDATE partition has a Cumulative Lift of 1.68 in the 10% quantile (depth of 10) meaning there are 1.68 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

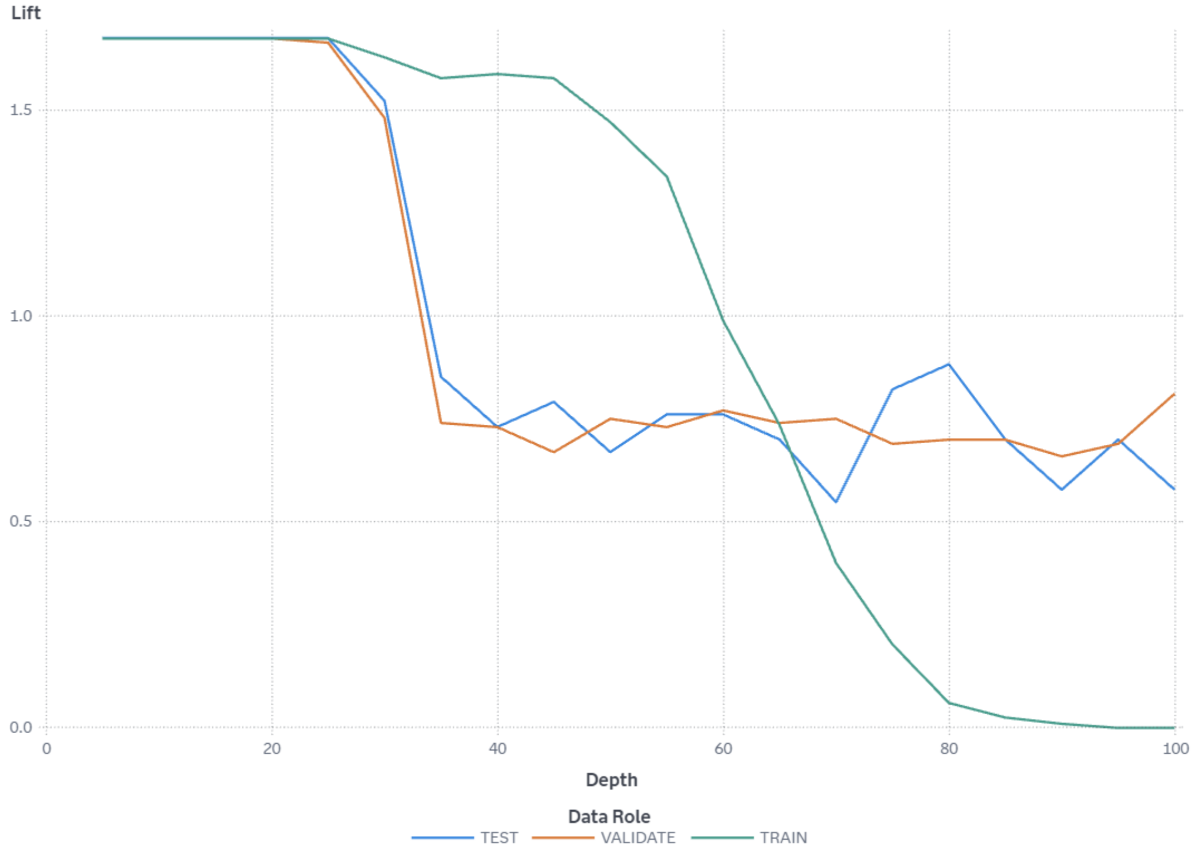
The TRAIN partition has a Cumulative Lift of 1.68 in the 10% quantile (depth of 10) meaning there are 1.68 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Cumulative Lift of 1.68 in the 10% quantile (depth of 10) meaning there are 1.68 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Cumulative lift is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target

Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative lift for a particular quantile is the ratio of the number of events across all quantiles up to and including the current quantile to the number of events that would be there at random, or equivalently, the ratio of the cumulative response percentage to the baseline response percentage. The cumulative lift at depth 10 includes the top 10% of the data, which is the first 2 quantiles, which would have 10% of the events at random. Thus, cumulative lift measures how much more likely it is to observe an event in the quantiles than by selecting observations at random.

Lift



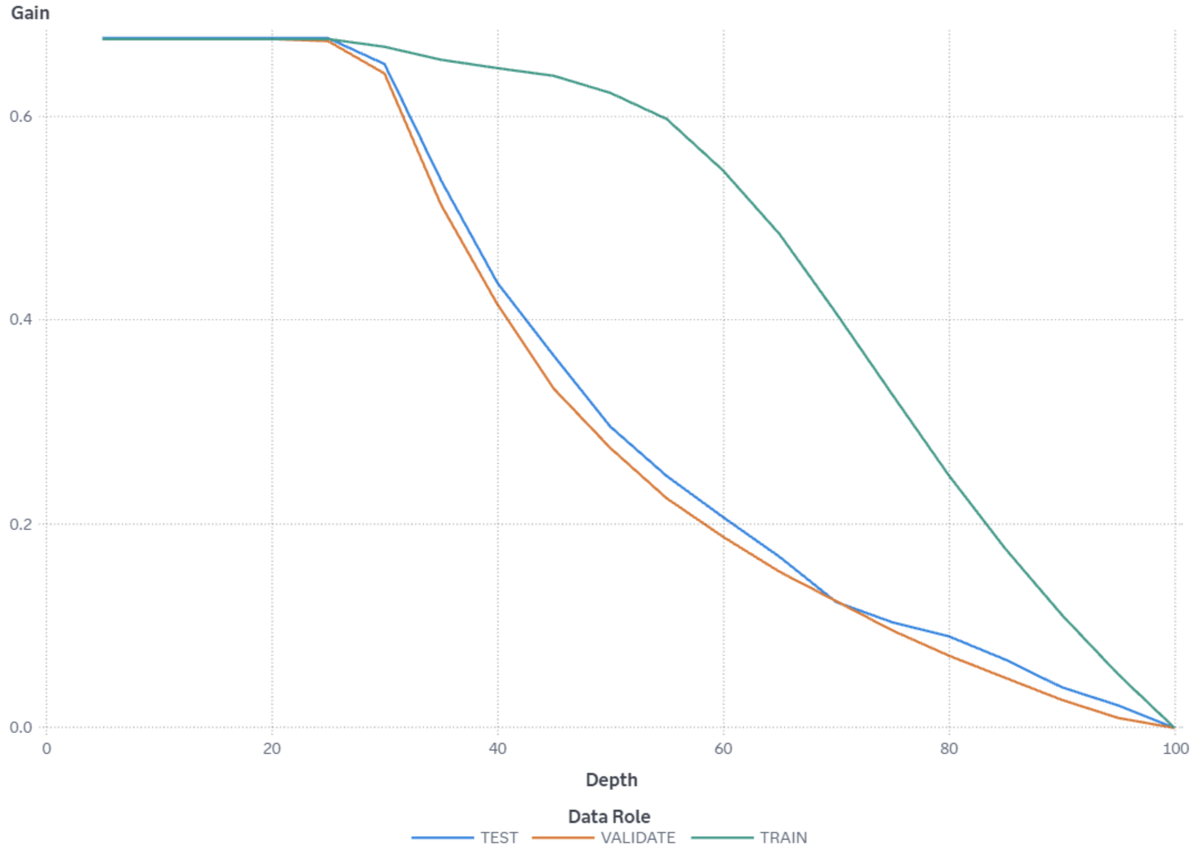
The VALIDATE partition has a Lift of 1.68 in the 5% quantile (depth of 5) meaning there are 1.68 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TRAIN partition has a Lift of 1.68 in the 5% quantile (depth of 5) meaning there are 1.68 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Lift of 1.68 in the 5% quantile (depth of 5) meaning there are 1.68 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Lift is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Lift is the ratio of the number of events in that quantile to the number of events that would be there at random, or equivalently, the ratio of the response percentage to the baseline response percentage. With 20 quantiles, it is expected that 5% of the events occur in each quantile. Thus, Lift measures how much more likely it is to observe an event in each quantile than by selecting observations at random.

Gain



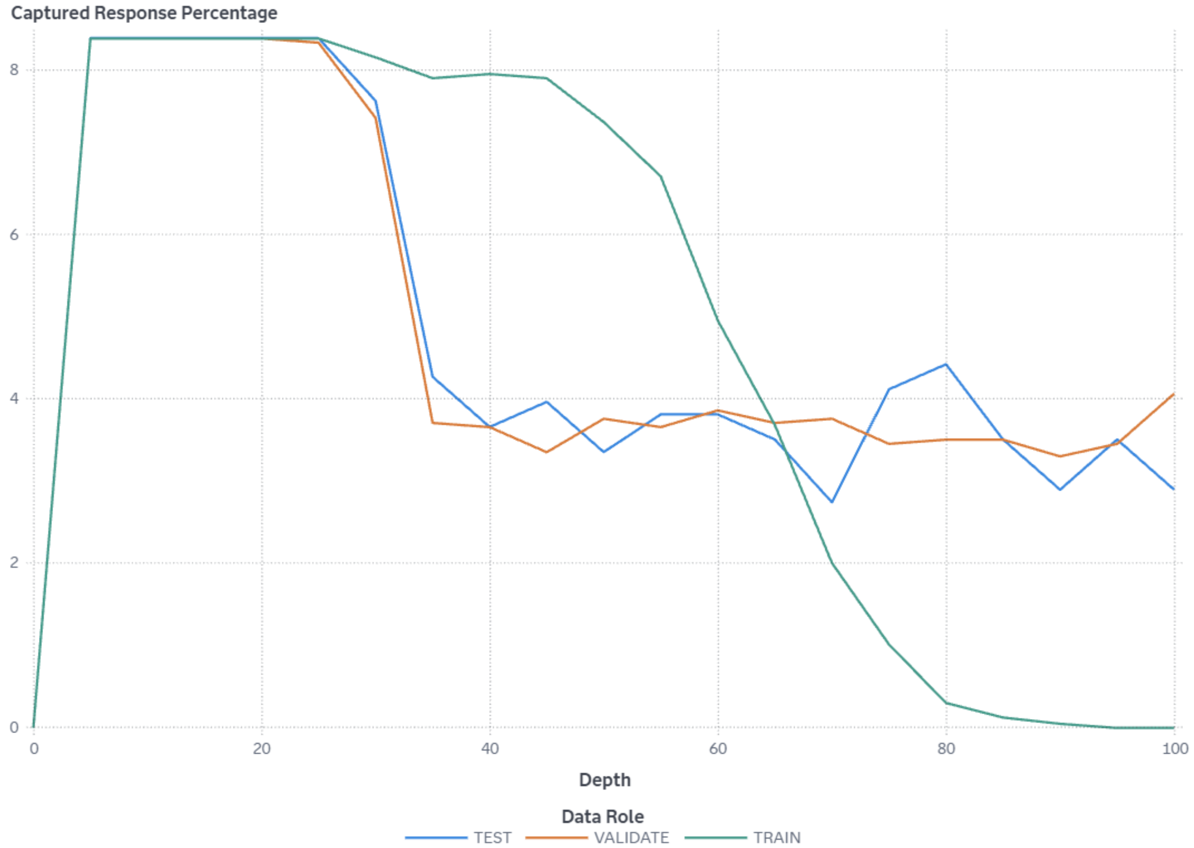
The VALIDATE partition has a Gain of 0.7 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

The TRAIN partition has a Gain of 0.7 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

The TEST partition has a Gain of 0.7 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

Gain is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Gain is a cumulative measure for the quantiles up to and including the current one and is calculated as (number of events in the quantiles) / (number of events expected by random) - 1. With 20 quantiles, it is expected that 5% of the events occur in each quantile. Note that the value of Gain is the same as the value of Cumulative Lift - 1. If the value of Gain is greater than 0, then your model is better at identifying events than using no model.

Captured Response Percentage



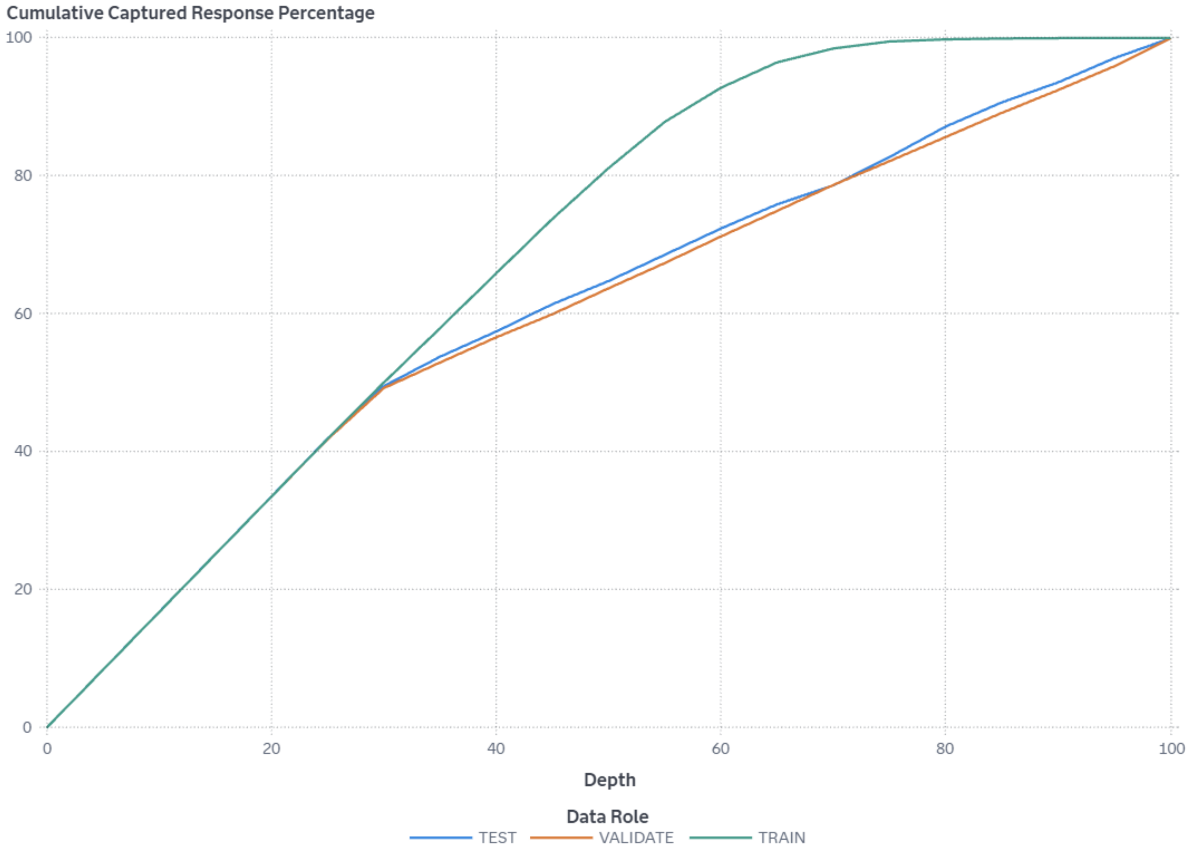
At the 5% quantile (depth of 5), the VALIDATE partition has a Captured response percentage of 8.4 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

At the 5% quantile (depth of 5), the TRAIN partition has a Captured response percentage of 8.4 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

At the 5% quantile (depth of 5), the TEST partition has a Captured response percentage of 8.4 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

Captured response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Captured response percentage is the percentage of the total number of events that are in that quantile. With no model, it is expected that 5% of the events are in each quantile.

Cumulative Captured Response Percentage



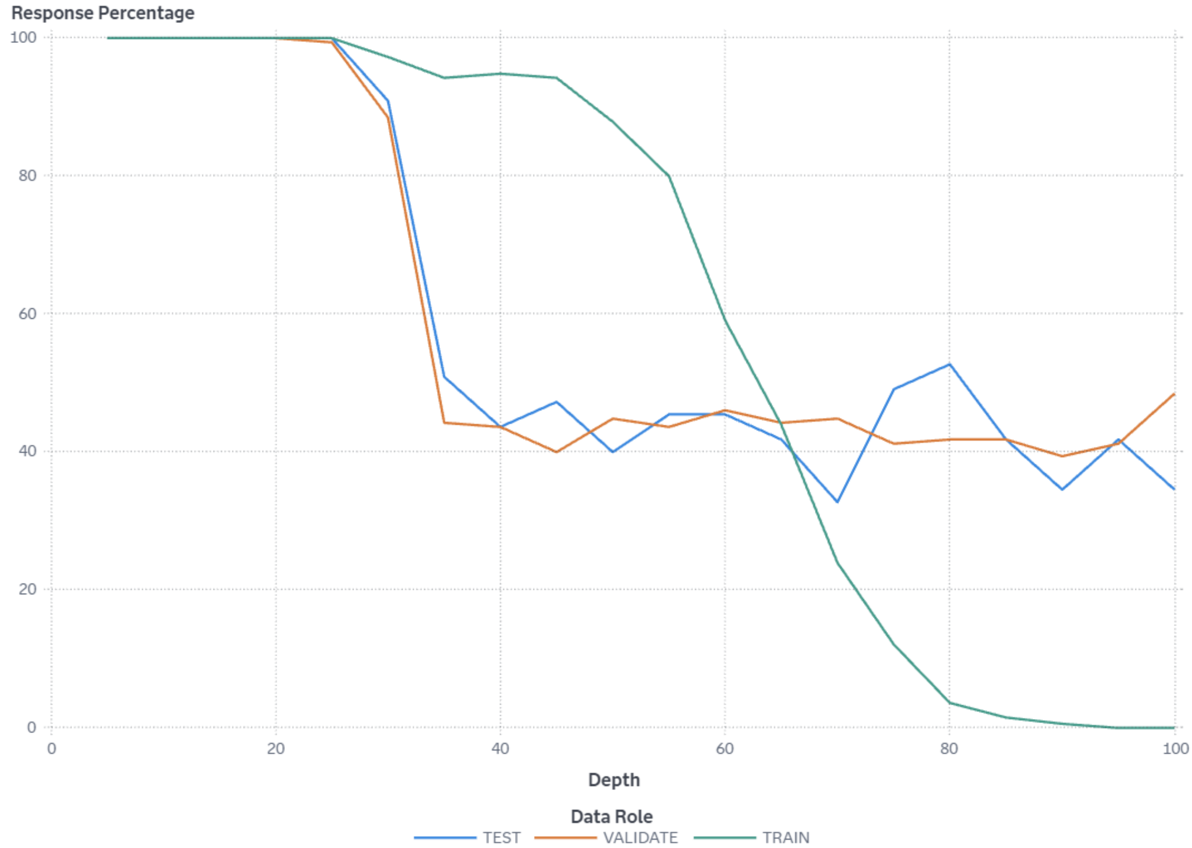
In the top 10% of the data (depth 10), the VALIDATE partition has a Cumulative captured response percentage of 16.8 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.76.

In the top 10% of the data (depth 10), the TRAIN partition has a Cumulative captured response percentage of 16.8 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.76.

In the top 10% of the data (depth 10), the TEST partition has a Cumulative captured response percentage of 16.8 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.77.

Cumulative captured response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative captured response percentage for a particular quantile is the percentage of the total number of events that are in the quantiles up to and including the current quantile. With no model, it is expected that 5% of the events are in each quantile, so the cumulative captured response percentage at depth 10 would be 10%.

Response Percentage



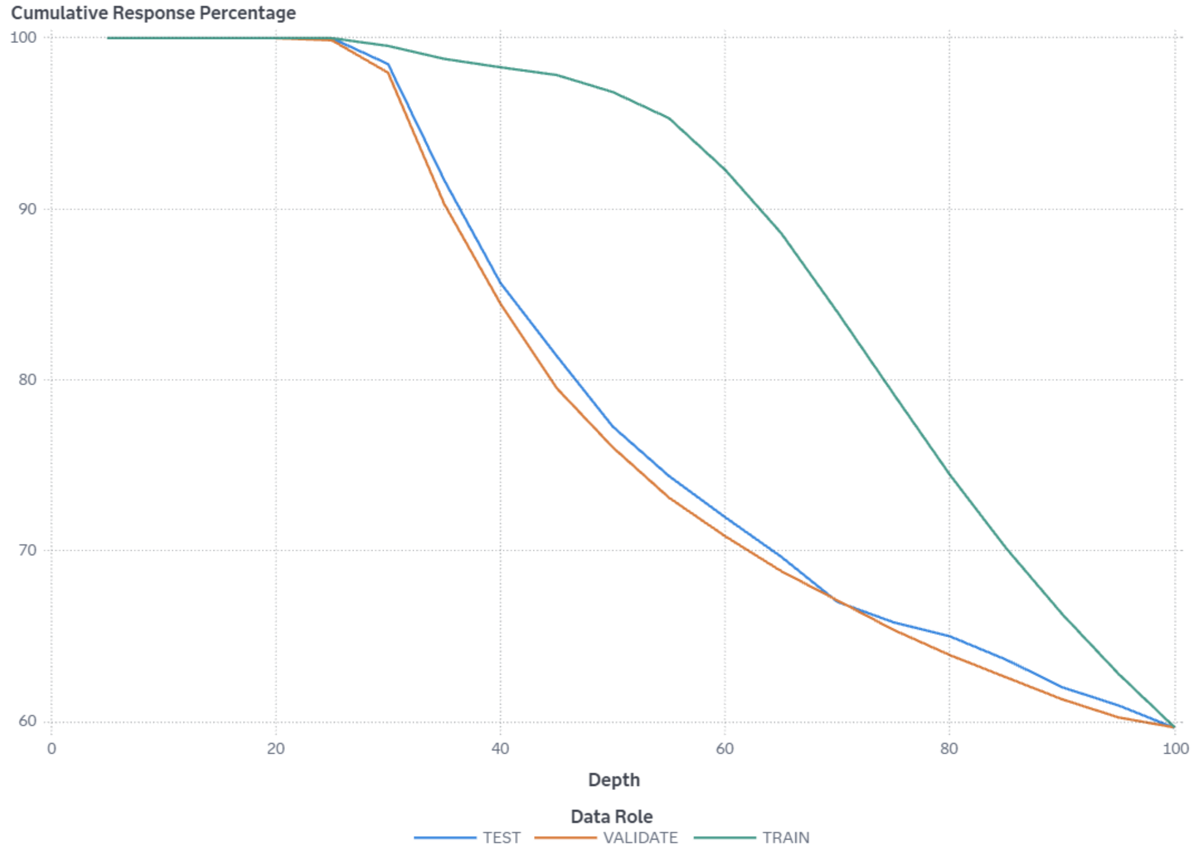
At the 5% quantile (depth of 5), the VALIDATE partition has a Response percentage of 100. The best possible value of Response percentage for this partition at depth 5 is 100.

At the 5% quantile (depth of 5), the TRAIN partition has a Response percentage of 100. The best possible value of Response percentage for this partition at depth 5 is 100.

At the 5% quantile (depth of 5), the TEST partition has a Response percentage of 100. The best possible value of Response percentage for this partition at depth 5 is 100.

Response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Response percentage is the percentage of observations that are events in that quantile. With no model, it is expected that the response percentage is constant across quantiles, 100*overall-event-rate. This is also called the baseline response percentage.

Cumulative Response Percentage



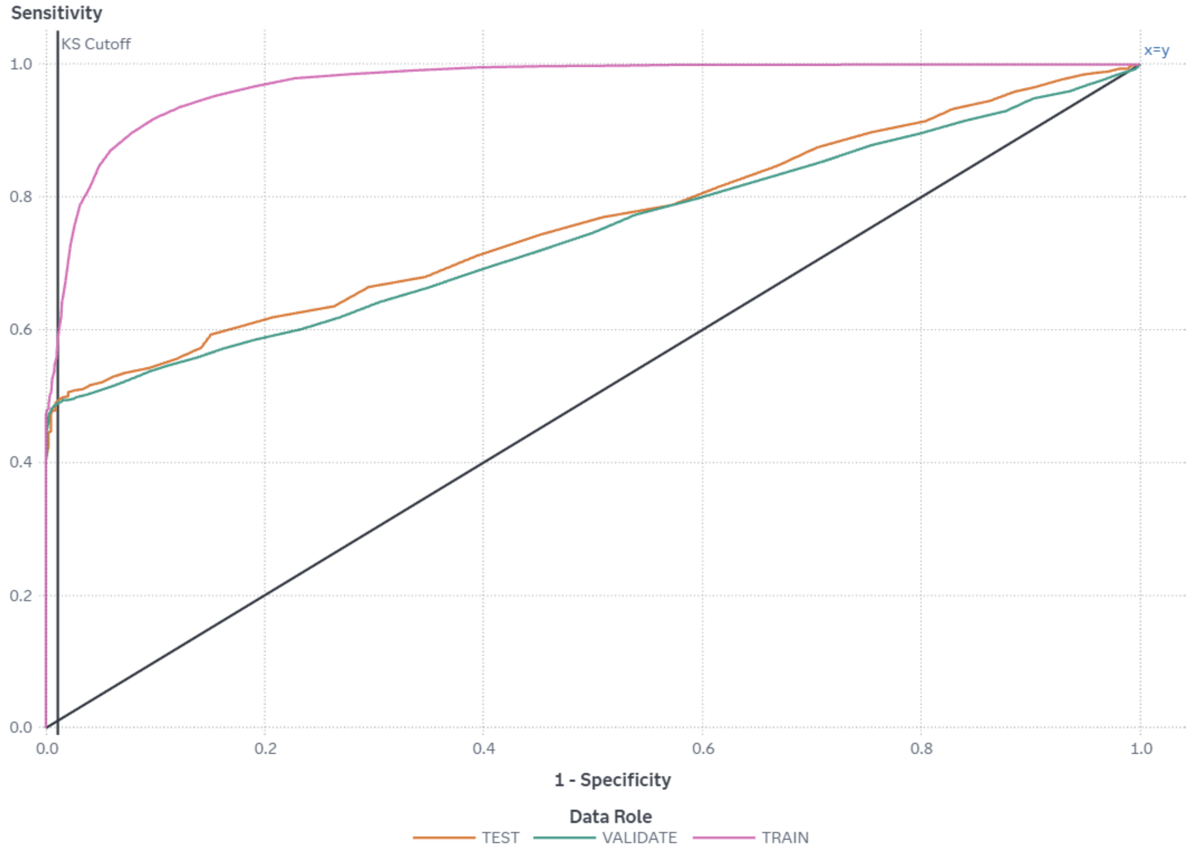
In the top 10% of the data (depth 10), the VALIDATE partition has a Cumulative response percentage of 100. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

In the top 10% of the data (depth 10), the TRAIN partition has a Cumulative response percentage of 100. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

In the top 10% of the data (depth 10), the TEST partition has a Cumulative response percentage of 100. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

Cumulative response percentage is calculated by sorting in descending order each partition of the data by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative response percentage for a particular quantile is the percentage of observations that are events in the quantiles up to and including the current quantile. With no model, it is expected that the response percentage is constant across quantiles, 100*overall-event-rate. This is also called the baseline response percentage.

ROC



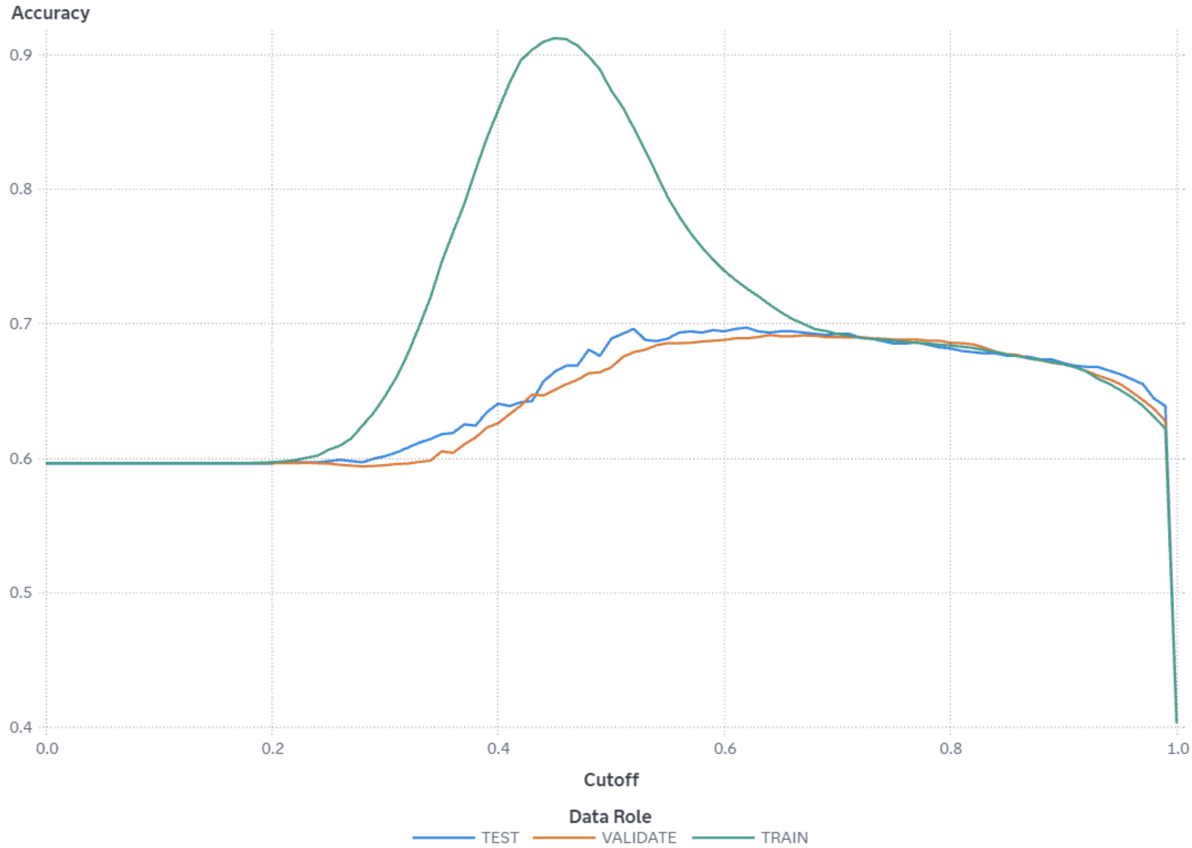
The ROC curve is a plot of sensitivity (the true positive rate) against 1-specificity (the false positive rate), which are both measures of classification based on the confusion matrix. These measures are calculated at various cutoff values. To help identify the best cutoff to use when scoring your data, the KS Cutoff reference line is drawn at the value of 1-specificity where the greatest difference between sensitivity and 1-specificity is observed for the VALIDATE partition. The KS Cutoff line is drawn at the cutoff value 0.67, where the 1-specificity value is 0.011 and the sensitivity value is 0.491.

Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether P_Reached_on_Time_Y_N1, which is the predicted probability of the event "1" for the target Reached.on.Time_Y.N, is greater than or equal to the cutoff value. When P_Reached_on_Time_Y_N1 is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event.

The confusion matrix for each cutoff value contains four cells that display the true positives for events that are correctly classified (TP), false positives for non-events that are classified as events (FP), false negatives for events that are classified as non-events (FN), and true negatives for non-events that are classified as non-events (TN). True negatives include non-event classifications that specify a different non-event. Sensitivity is calculated as TP / (TP + FN). Specificity, the true negative rate, is calculated as TN / (TN + FP), so 1-specificity is FP / (TN + FP). The values of sensitivity and 1-specificity are plotted at each cutoff value.

A ROC curve that rapidly approaches the upper-left corner of the graph, where the difference between sensitivity and 1-specificity is the greatest, indicates a more accurate model. A diagonal line where sensitivity = 1-specificity indicates a random model.

Accuracy



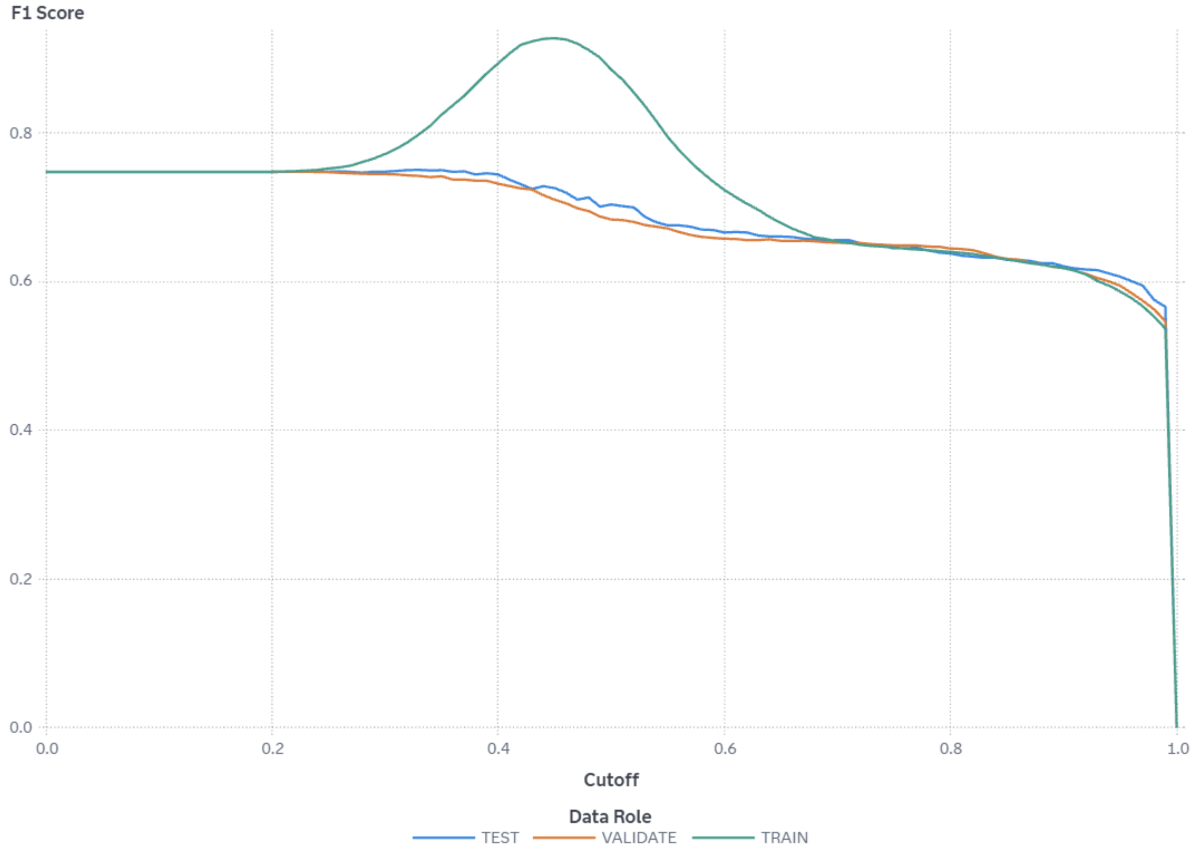
For this model, the accuracy in the TEST partition at the cutoff of 0.5 is 0.689.

For this model, the accuracy in the TRAIN partition at the cutoff of 0.5 is 0.873.

For this model, the accuracy in the VALIDATE partition at the cutoff of 0.5 is 0.668.

Accuracy is the proportion of observations that are correctly classified as either an event or non-event, calculated at various cutoff values. Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether P_Reached_on_Time_Y_N1, which is the predicted probability of the event "1" for the target Reached.on.Time_Y.N, is greater than or equal to the cutoff value. When P_Reached_on_Time_Y_N1 is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event. When the predicted classification and the actual classification are both events (true positives) or both non-events (true negatives), the observation is correctly classified. If the predicted classification and actual classification disagree, then the observation is incorrectly classified. Accuracy is calculated as (true positives + true negatives) / (total observations).

F1 Score



For this model, the F1 score in the TEST partition at the cutoff of 0.5 is 0.704.

For this model, the F1 score in the TRAIN partition at the cutoff of 0.5 is 0.885.

For this model, the F1 score in the VALIDATE partition at the cutoff of 0.5 is 0.683.

The F1 score combines the measures of precision and recall (or sensitivity), which are measures of classification based on the confusion matrix that are calculated at various cutoff values. Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether P_Reached_on_Time_Y_N1, which is the predicted probability of the event "1" for the target Reached.on.Time_Y.N, is greater than or equal to the cutoff value. When P_Reached_on_Time_Y_N1 is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event.

The confusion matrix for each cutoff value contains four cells that display the true positives for events that are correctly classified (TP), false positives for non-events that are classified as events (FP), false negatives for events that are classified as non-events (FN), and true negatives for non-events that are classified as non-events (TN). True negatives include non-event classifications that specify a different non-event.

Precision is calculated as TP / (TP + FP), and recall (or sensitivity) is calculated as TP / (TP + FN). The F1 score is calculated as 2*Precision*Recall / (Precision + Recall), which is the harmonic mean of Precision and Recall. Larger F1 scores indicate a more accurate model.

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
Reached.on.Time_Y.N	TEST	2	2
Reached.on.Time_Y.N	TRAIN	1	1
Reached.on.Time_Y.N	VALIDATE	0	0

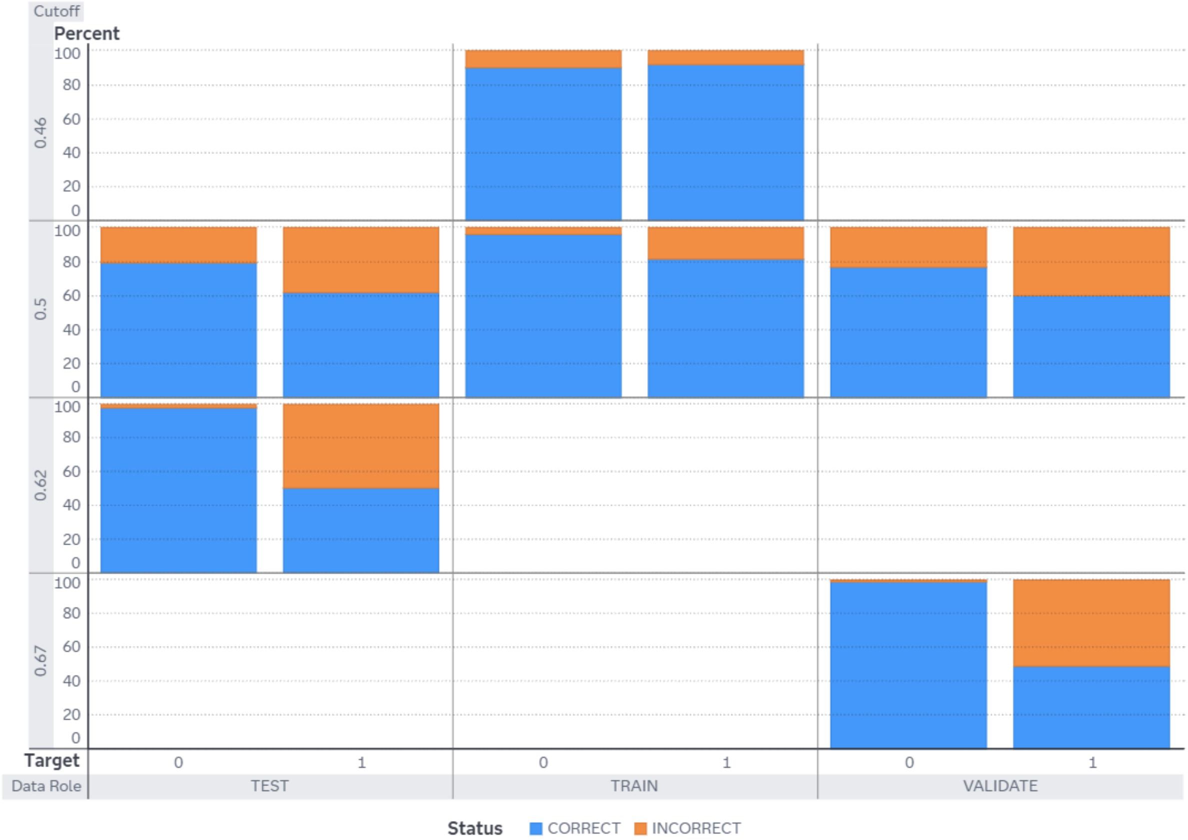
Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,100	0.1771	1,100	0.4208
6,599	0.1295	6,599	0.3599
3,300	0.1804	3,300	0.4248

Misclassification Rate	Multi-Class Log Loss	KS (Youden)	Area Under ROC
0.3109	0.4975	0.4858	0.7602
0.1267	0.3983	0.8195	0.9715
0.3321	0.5040	0.4793	0.7433

Gini Coefficient	Gamma	Tau	KS Cutoff
0.5203	0.5301	0.2507	0.6200
0.9430	0.9481	0.4539	0.4600
0.4865	0.4960	0.2342	0.6700

KS at User-Specified Cutoff	Misclassification Rate at KS Cutoff (Event)	Misclassification Rate (Event)
0.4117	0.3027	0.3109
0.7747	0.0885	0.1267
0.3679	0.3085	0.3321

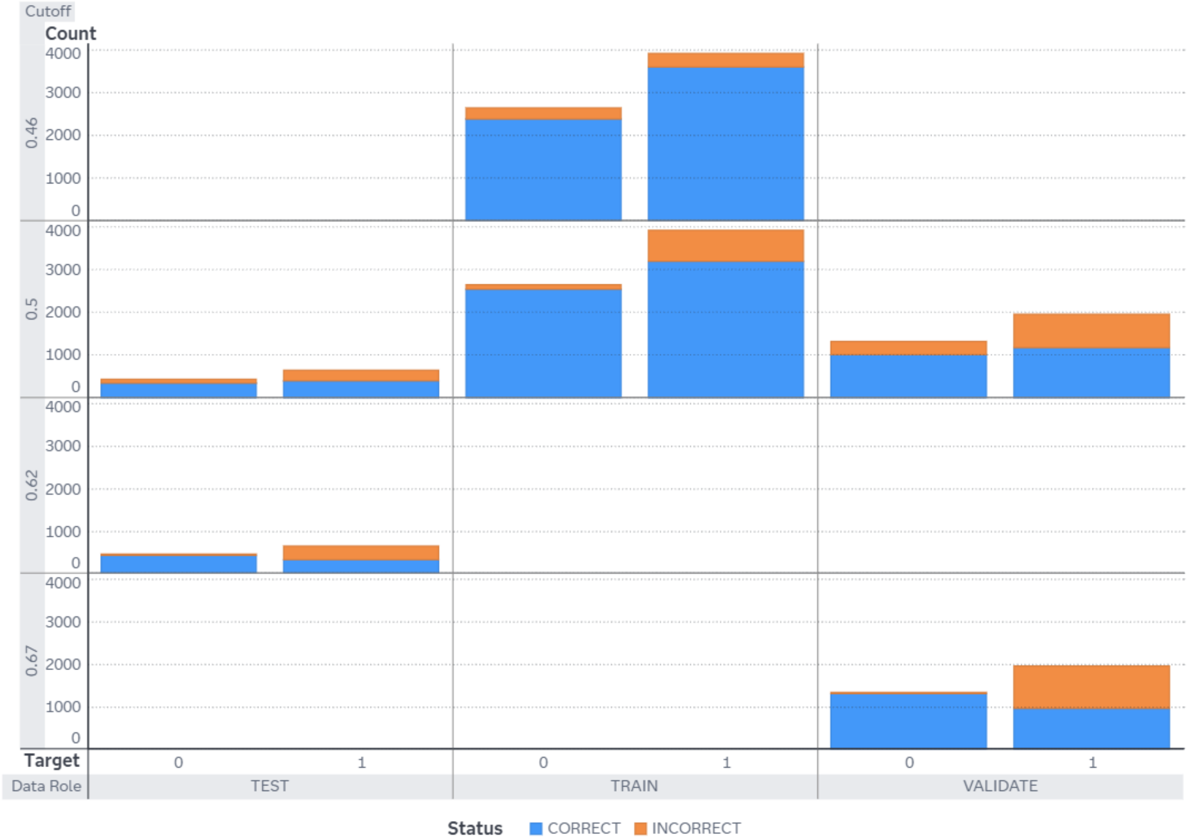
Percentage Plot



The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0.46 (TRAIN), 0.67 (VALIDATE), 0.62 (TEST).

For this data, for the bar corresponding to the event level of Reached.on.Time_Y.N, "1", the segment of the bar colored as "CORRECT" corresponds to true positives.

Count Plot



The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0.46 (TRAIN), 0.67 (VALIDATE), 0.62 (TEST).

For this data, for the bar corresponding to the event level of Reached.on.Time_Y.N, "1", the segment of the bar colored as "CORRECT" corresponds to true positives.

Table

Cutoff	Cutoff Source	Target Name	Response
0.4600	KS	Reached.on.Time_Y.N	CORRECT
0.4600	KS	Reached.on.Time_Y.N	INCORRECT
0.4600	KS	Reached.on.Time_Y.N	CORRECT
0.4600	KS	Reached.on.Time_Y.N	INCORRECT
0.5000	Default	Reached.on.Time_Y.N	CORRECT
0.5000	Default	Reached.on.Time_Y.N	INCORRECT
0.5000	Default	Reached.on.Time_Y.N	CORRECT
0.5000	Default	Reached.on.Time_Y.N	INCORRECT
0.6200	KS	Reached.on.Time_Y.N	CORRECT
0.6200	KS	Reached.on.Time_Y.N	INCORRECT
0.6200	KS	Reached.on.Time_Y.N	CORRECT
0.6200	KS	Reached.on.Time_Y.N	INCORRECT
0.6700	KS	Reached.on.Time_Y.N	CORRECT
0.6700	KS	Reached.on.Time_Y.N	INCORRECT
0.6700	KS	Reached.on.Time_Y.N	CORRECT
0.6700	KS	Reached.on.Time_Y.N	INCORRECT

Event	Value	Training Frequency	Validation Frequency

Event	Value	Training Frequency	Validation Frequency
1	True Positive	3,618	
1	False Negative	320	
0	True Negative	2,397	
0	False Positive	264	
1	True Positive	3,209	1,183
1	False Negative	729	786
0	True Negative	2,554	1,021
0	False Positive	107	310
1	True Positive		
1	False Negative		
0	True Negative		
0	False Positive		
1	True Positive		966
1	False Negative		1,003
0	True Negative		1,316
0	False Positive		15

Test Frequency	Training Percentage	Validation Percentage	Test Percentage
	91.8740		
	8.1260		
	90.0789		
	9.9211		
406	81.4881	60.0813	61.8902
250	18.5119	39.9187	38.1098
352	95.9790	76.7092	79.2793
92	4.0210	23.2908	20.7207
332			50.6098
324			49.3902
435			97.9730
9			2.0270

Test Frequency	Training Percentage	Validation Percentage	Test Percentage
		49.0604	
		50.9396	
		98.8730	
		1.1270	

Properties

Property Name	Property Value
atAppendLookup	false
atCreateHistory	false
atHistoryLibUri	
atHistoryTblName	
atLeaveAutotuneOn	false
atLookupTableUri	
atMaxBayes	100
atMaxEval	50
atMaxIter	5
atMaxTime	60
atObjectiveInt	ASE
atObjectiveNom	KS
atPopSize	10
atSampleSize	50
atSearchMethod	GA
atTrainProp	0.7000
atUpdateProperties	false
atUseLookup	false
atValidFold	5
atValidMethod	PARTITION
atValidProp	0.3000
atintervalBins	true
atintervalBinsInit	50
atintervalBinsLB	20
atintervalBinsUB	100
atleafSize	false
atleafSizeInit	5
atleafSizeLB	1
atleafSizeUB	100

Property Name	Property Value
atmaxDepth	true
atmaxDepthInit	20
atmaxDepthLB	1
atmaxDepthUB	29
atmaxTrees	true
atmaxTreesInit	100
atmaxTreesLB	20
atmaxTreesUB	150
attrainFraction	true
attrainFractionInit	0.6000
attrainFractionLB	0.1000
attrainFractionUB	0.9000
atvarsToTry	true
atvarsToTryInit	100
atvarsToTryLB	1
atvarsToTryUB	100
autotune_enabled	false
binaryProbCutoff	0.5000
codeLocation	mlearning
criterionMethod	IGR
dataMiningVersion	V2025.08
defaultVarsPerTree	true
exactPctlLift	true
explainFidelity	false
explainInfo	false
fullDatasetReconstitution	false
iCriterionMethod	VARIANCE
icePlots	false
intBinMethod	QUANTILE
intervalBins	50

Property Name	Property Value
leafProp	0.0001
leafSize	5
leafSpec	COUNT
loh	0
maxBranch	2
maxCategories	128
maxDepth	20
maxNumShapVars	20
maxTrees	100
minUseInSearch	1
missingValue	USEINSEARCH
nBins	50
pdNumImportantInputs	5
pdObsSamples	1,000
pdPlots	false
performKernelShap	false
performLime	false
performVI	false
seed	12,345
seedId	12,345
specifyRows	RANDOM
templateRevision	4
train	true
trainFraction	0.6000
truncateLI	5
truncateUI	95
userProbCutoff	false
varsToTry	100
voteMethod	PROBABILITY

Output

The SAS System

The FOREST Procedure

Model Information	
Number of Trees	100
Number of Variables Per Split	4
Seed	12345
Bootstrap Percentage	60
Number of Bins	50
Number of Input Variables	10
Maximum Number of Tree Nodes	803
Minimum Number of Tree Nodes	447
Maximum Number of Branches	2
Minimum Number of Branches	2
Maximum Depth	20
Minimum Depth	20
Maximum Number of Leaves	402
Minimum Number of Leaves	224
Maximum Leaf Size	936
Minimum Leaf Size	5
OOB Misclassification Rate	0.33792999
Average Number of Leaves	323.6

	Training	Validation	Test	Total
Number of Observations Read	6599	3300	1100	10999
Number of Observations Used	6599	3300	1100	10999

Variable Importance

Variable	Importance	Std Dev Importance	Relative Importance
Discount_offered	325.58	129.35	1.0000
Weight_in_gms	245.66	117.66	0.7545
Cost_of_the_Product	132.78	30.3922	0.4078
Prior_purchases	91.2513	38.5608	0.2803
Warehouse_block	64.9953	13.9537	0.1996
Customer_rating	63.4539	14.9706	0.1949
Customer_care_calls	59.8808	17.9286	0.1839
Mode_of_Shipment	26.2470	7.6626	0.0806
Product_importance	24.9246	7.5247	0.0766
Gender	17.3740	6.9544	0.0534

Fit Statistics

Number of Trees	OOB Average Square Error	Training Average Square Error	Validation Average Square Error	Test Average Square Error	OOB Misclassification Rate	Training Misclassification Rate	Validation Misclassification Rate	Test Misclassification Rate	OOB Log Loss	Training Log Loss	Validation Log Loss	Test Log Loss
1	0.265	0.193	0.256	0.269	0.363	0.273	0.344	0.347	2.993	1.769	2.973	3.343
2	0.238	0.162	0.219	0.213	0.357	0.252	0.352	0.326	2.006	0.602	1.001	1.054
3	0.227	0.153	0.206	0.199	0.354	0.238	0.345	0.322	1.505	0.465	0.632	0.613
4	0.221	0.147	0.199	0.194	0.359	0.232	0.352	0.325	1.211	0.434	0.570	0.576
5	0.217	0.142	0.197	0.190	0.359	0.213	0.349	0.331	1.006	0.421	0.563	0.546
6	0.213	0.139	0.196	0.189	0.359	0.198	0.349	0.321	0.848	0.414	0.553	0.544
7	0.208	0.139	0.194	0.188	0.358	0.195	0.352	0.332	0.725	0.413	0.541	0.539
8	0.205	0.137	0.192	0.187	0.356	0.193	0.345	0.340	0.673	0.411	0.536	0.536
9	0.201	0.136	0.191	0.185	0.353	0.184	0.350	0.343	0.620	0.408	0.534	0.532
10	0.199	0.135	0.190	0.185	0.354	0.179	0.351	0.333	0.597	0.406	0.532	0.533
11	0.198	0.134	0.190	0.185	0.351	0.175	0.347	0.336	0.574	0.406	0.531	0.533
12	0.195	0.133	0.189	0.184	0.346	0.175	0.341	0.334	0.555	0.404	0.529	0.531
13	0.194	0.133	0.188	0.184	0.346	0.166	0.335	0.335	0.553	0.403	0.527	0.530
14	0.193	0.133	0.187	0.183	0.344	0.161	0.330	0.327	0.545	0.403	0.524	0.529
15	0.192	0.132	0.186	0.183	0.343	0.159	0.333	0.323	0.542	0.402	0.522	0.528
16	0.192	0.132	0.185	0.183	0.344	0.158	0.332	0.330	0.534	0.402	0.521	0.527
17	0.191	0.132	0.185	0.182	0.345	0.154	0.333	0.325	0.532	0.403	0.520	0.526
18	0.191	0.132	0.184	0.182	0.345	0.154	0.334	0.324	0.529	0.402	0.519	0.525
19	0.191	0.132	0.184	0.182	0.345	0.157	0.334	0.332	0.528	0.402	0.513	0.524
20	0.190	0.132	0.184	0.182	0.345	0.156	0.333	0.319	0.527	0.402	0.512	0.524
21	0.190	0.132	0.183	0.181	0.345	0.156	0.328	0.327	0.526	0.402	0.511	0.524
22	0.189	0.132	0.183	0.181	0.342	0.152	0.328	0.324	0.525	0.402	0.510	0.522
23	0.189	0.132	0.183	0.180	0.345	0.151	0.331	0.324	0.524	0.401	0.509	0.522
24	0.189	0.132	0.182	0.181	0.345	0.150	0.331	0.323	0.524	0.402	0.508	0.522
25	0.189	0.132	0.182	0.181	0.345	0.147	0.332	0.326	0.523	0.402	0.508	0.523
26	0.189	0.131	0.182	0.180	0.348	0.147	0.331	0.328	0.523	0.401	0.508	0.522
27	0.189	0.131	0.182	0.180	0.348	0.146	0.329	0.325	0.523	0.401	0.507	0.505
28	0.188	0.131	0.182	0.180	0.347	0.144	0.327	0.327	0.522	0.401	0.507	0.505
29	0.188	0.131	0.182	0.180	0.346	0.144	0.329	0.329	0.522	0.401	0.508	0.505
30	0.188	0.131	0.182	0.180	0.345	0.141	0.326	0.327	0.522	0.400	0.508	0.505
31	0.188	0.131	0.182	0.180	0.345	0.139	0.331	0.328	0.521	0.400	0.507	0.505
32	0.188	0.130	0.182	0.180	0.345	0.141	0.331	0.329	0.520	0.399	0.508	0.505
33	0.188	0.130	0.182	0.180	0.344	0.138	0.333	0.327	0.521	0.399	0.508	0.505
34	0.187	0.130	0.182	0.180	0.346	0.138	0.337	0.325	0.520	0.400	0.507	0.504
35	0.187	0.130	0.182	0.180	0.344	0.137	0.332	0.319	0.520	0.399	0.508	0.504
36	0.187	0.130	0.182	0.180	0.345	0.136	0.332	0.317	0.520	0.399	0.507	0.503
37	0.187	0.130	0.182	0.179	0.344	0.136	0.331	0.315	0.519	0.399	0.507	0.501
38	0.187	0.130	0.182	0.179	0.344	0.136	0.330	0.314	0.519	0.399	0.507	0.501
39	0.187	0.130	0.181	0.178	0.345	0.137	0.328	0.315	0.520	0.399	0.507	0.501
40	0.187	0.130	0.181	0.179	0.344	0.136	0.331	0.316	0.519	0.399	0.506	0.502
41	0.187	0.130	0.182	0.179	0.342	0.137	0.331	0.314	0.519	0.399	0.507	0.502
42	0.187	0.130	0.182	0.179	0.343	0.137	0.328	0.319	0.519	0.399	0.507	0.501
43	0.187	0.130	0.182	0.179	0.342	0.138	0.329	0.321	0.518	0.399	0.507	0.502
44	0.187	0.130	0.182	0.179	0.343	0.135	0.328	0.314	0.518	0.399	0.507	0.501
45	0.186	0.130	0.182	0.178	0.342	0.135	0.332	0.311	0.518	0.399	0.507	0.500
46	0.186	0.130	0.182	0.178	0.344	0.136	0.328	0.311	0.518	0.399	0.507	0.500
47	0.186	0.130	0.181	0.178	0.343	0.136	0.328	0.316	0.518	0.399	0.506	0.499
48	0.186	0.130	0.181	0.178	0.342	0.137	0.327	0.314	0.518	0.399	0.506	0.500
49	0.186	0.130	0.181	0.178	0.341	0.134	0.330	0.313	0.517	0.399	0.506	0.500
50	0.186	0.130	0.181	0.178	0.341	0.135	0.328	0.316	0.517	0.398	0.506	0.499
51	0.186	0.130	0.181	0.178	0.340	0.133	0.331	0.315	0.517	0.398	0.506	0.499
52	0.186	0.130	0.181	0.178	0.339	0.134	0.330	0.317	0.517	0.398	0.506	0.499
53	0.186	0.130	0.181	0.178	0.341	0.136	0.331	0.319	0.517	0.399	0.506	0.499
54	0.186	0.130	0.181	0.178	0.342	0.137	0.331	0.322	0.517	0.399	0.506	0.499
55	0.186	0.130	0.181	0.178	0.342	0.135	0.330	0.326	0.517	0.398	0.506	0.499
56	0.186	0.130	0.181	0.178	0.342	0.136	0.329	0.321	0.517	0.399	0.506	0.499
57	0.186	0.130	0.181	0.178	0.341	0.137	0.328	0.323	0.517	0.398	0.506	0.499
58	0.186	0.130	0.181	0.178	0.342	0.137	0.328	0.322	0.516	0.398	0.506	0.499
59	0.186	0.130	0.181	0.178	0.341	0.137	0.329	0.323	0.516	0.399	0.506	0.499
60	0.186	0.130	0.181	0.178	0.339	0.137	0.329	0.320	0.517	0.399	0.506	0.499
61	0.186	0.130	0.181	0.178	0.339	0.135	0.330	0.320	0.517	0.399	0.505	0.499
62	0.186	0.130	0.181	0.178	0.338	0.135	0.326	0.318	0.517	0.399	0.505	0.499
63	0.186	0.130	0.181	0.177	0.339	0.136	0.328	0.316	0.517	0.399	0.505	0.498
64	0.186	0.130	0.181	0.177	0.339	0.135	0.324	0.315	0.517	0.399	0.505	0.498
65	0.186	0.130	0.181	0.177	0.338	0.134	0.325	0.316	0.517	0.399	0.505	0.498
66	0.186	0.130	0.181	0.177	0.337	0.132	0.325	0.309	0.516	0.398	0.505	0.498
67	0.186	0.130	0.181	0.177	0.339	0.132	0.328	0.308	0.516	0.398	0.504	0.497
68	0.186	0.130	0.181	0.177	0.338	0.132	0.325	0.313	0.516	0.398	0.504	0.497
69	0.186	0.130	0.181	0.177	0.338	0.129	0.325	0.312	0.516	0.398	0.504	0.497
70	0.186	0.130	0.181	0.177	0.337	0.128	0.327	0.314	0.516	0.398	0.504	0.498
71	0.186	0.130	0.181	0.177	0.338	0.127	0.329	0.315	0.516	0.398	0.504	0.498
72	0.186	0.130	0.181	0.177	0.338	0.129	0.329	0.313	0.516	0.398	0.504	0.497
73	0.186	0.130	0.181	0.177	0.337	0.127	0.330	0.316	0.516	0.398	0.504	0.497
74	0.186	0.130	0.181	0.177	0.339	0.128	0.329	0.315	0.516	0.399	0.504	0.497
75	0.186	0.130	0.181	0.177	0.338	0.129	0.328	0.318	0.516	0.399	0.504	0.497
76	0.185	0.130	0.180	0.177	0.339	0.128	0.328	0.317	0.516	0.398	0.504	0.498
77	0.185	0.130	0.181	0.177	0.339	0.128	0.329	0.317	0.516	0.398	0.504	0.498
78	0.185	0.130	0.180	0.177	0.340	0.129	0.328	0.318	0.516	0.398	0.504	0.498
79	0.185	0.130	0.180	0.177	0.339	0.129	0.330	0.315	0.516	0.399	0.504	0.497
80	0.185	0.130	0.180	0.177	0.338	0.129	0.328	0.313	0.516	0.398	0.504	0.497
81	0.185	0.130	0.180	0.177	0.340	0.129	0.328	0.312	0.516	0.398	0.504	0.497
82	0.185	0.130	0.180	0.177	0.340	0.128	0.329	0.314	0.516	0.399	0.504	0.497
83	0.185	0.130	0.180	0.177	0.340	0.130	0.331	0.315	0.516	0.399	0.504	0.497
84	0.185	0.130	0.180	0.177	0.340	0.128	0.330	0.315	0.516	0.399	0.504	0.498
85	0.185	0.130	0.180	0.177	0.339	0.127	0.329	0.316	0.516	0.399	0.504	0.497
86	0.185	0.130	0.180	0.177	0.338	0.127	0.330	0.317	0.516	0.398	0.504	0.498
87	0.185	0.130	0.180	0.177	0.338	0.127	0.330	0.316	0.515	0.399	0.504	0.497
88	0.185	0.130	0.181	0.177	0.338	0.127	0.331	0.315	0.515	0.399	0.504	0.497
89	0.185	0.130	0.181	0.177	0.339	0.126	0.334	0.318	0.515	0.398	0.504	0.498
90	0.185	0.130	0.181	0.177	0.339	0.125	0.335	0.321	0.515	0.398	0.504	0.497
91	0.185	0.130	0.181	0.177	0.338	0.127	0.335	0.323	0.515	0.398	0.504	0.498
92	0.185	0.129	0.181	0.177	0.337	0.127	0.334	0.322	0.515	0.398	0.504	0.498
93	0.185	0.130	0.180	0.177	0.337	0.125	0.332	0.321	0.515	0.398	0.504	0.498
94	0.185	0.130	0.180	0.177	0.337	0.127	0.333	0.319	0.515	0.398	0.504	0.498
95	0.185	0.130	0.181	0.177	0.338	0.127	0.332	0.321	0.515	0.398	0.504	0.498
96	0.185	0.130	0.181	0.177	0.338	0.126	0.331	0.318	0.515	0.398	0.504	0.498
97	0.185	0.130	0.181	0.177	0.339	0.126	0.331	0.315	0.515	0.398	0.504	0.498
98	0.185	0.129	0.180	0.177	0.337	0.126	0.331	0.315	0.515	0.398	0.504	0.498
99	0.185	0.130	0.180	0.177	0.339	0.127	0.333	0.316	0.515	0.398	0.504	0.498
100	0.185	0.130	0.180	0.177	0.338	0.127	0.332	0.311	0.514	0.398	0.504	0.498

Output CAS Tables

CAS Library	Name	Number of Rows	Number of Columns
Analytics_Project_863d0c53f2ae-4559-81c5-6981213c1802	_4OB5DQDP04E4GQSXQ4XUQ355L_AST	1	2

Predicted Probability Variables

Reached.on.Time_Y.N	Variable
1	P_Reached_on_Time_Y_N1
0	P_Reached_on_Time_Y_N0

Predicted Target Variable

Level Index	Variable
	I_Reached_on_Time_Y_N